# EXPERIMENT – 6

## Snort IDS/IPS to Detect and Prevent Real-Time Threats Using TryHackMe Platform

### Aim

To understand and demonstrate the working of Snort as an Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) by detecting real-time network threats using the TryHackMe platform.

### Tools / Platform Required

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Snort IDS/IPS Tool (Pre-installed in AttackBox)

### Theory (Brief)

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) are essential components of network security used to monitor, detect, and prevent malicious activities.
Snort is an open-source, signature-based IDS/IPS capable of real-time traffic analysis and packet inspection.
TryHackMe provides a safe and legal environment to practice IDS/IPS concepts by simulating real-world attacks and defenses without impacting live systems.

### Steps to Perform the Experiment

### Step 1: Login to TryHackMe

1. Open browser and go to: https://tryhackme.com
2. Login using registered credentials.

### Step 2: Open Snort Room

1. In the search bar, type **Snort**.
2. Select a room related to **Snort IDS/IPS**.
3. Click **Join Room**.

### Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

### Step 4: Verify Snort Installation

1. In the terminal, check Snort version:
2. snort --version
3. Ensure Snort is successfully installed.

**Step 5: Run Snort in IDS Mode**

1. Start Snort in console mode:
2. sudo snort -A console -q -c /etc/snort/snort.conf -i eth0
3. Snort begins monitoring network traffic in real time.

**Step 6: Detect Network Threats**

1. Perform network scanning or simulated attacks as instructed in the TryHackMe room.
2. Observe alerts generated by Snort.
3. Analyze alerts indicating suspicious or malicious activity.

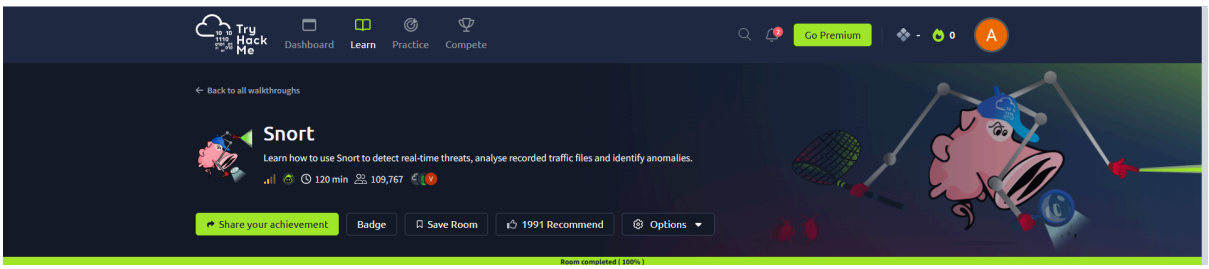
**Step 7: Configure Snort as IPS (Optional)**

1. Run Snort in inline (IPS) mode:
2. sudo snort -Q --daq afpacket -c /etc/snort/snort.conf -i eth0
3. Configure rules to drop malicious packets.
4. Observe blocked traffic in real time.

**Step 8: Analyze Alerts**

1. Review generated alerts on the terminal.
2. Identify attack types such as port scanning, brute force, or suspicious packets.

**Output**




```
labex:project/ $ snort --version

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.15.1 GRE (Build 15125)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reser
ved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.10.1 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.2.11
```

```
Run time for packet processing was 69.555474 seconds
Snort processed 7542 packets.
Snort ran for 0 days 0 hours 1 minutes 9 seconds
   Pkts/min:            7542
   Pkts/sec:             109
==================================================================================
Memory usage summary:
  Total non-mmapped bytes (arena):       790528
  Bytes in mapped regions (hblkhd):      21590016
  Total allocated space (uordblks):      684512
  Total free space (fordblks):           106016
  Topmost releasable block (keepcost):   103424
==================================================================================
Packet I/O Totals:
   Received:            7656
   Analyzed:            7542 ( 98.511%)
    Dropped:               0 (  0.000%)
   Filtered:               0 (  0.000%)
Outstanding:             114 (  1.489%)
   Injected:               0
```

```
==============================================================
Run time for packet processing was 127.999614 seconds
Snort processed 81 packets.
Snort ran for 0 days 0 hours 2 minutes 7 seconds
   Pkts/min:            40
   Pkts/sec:             0
==============================================================
Memory usage summary:
  Total non-mmapped bytes (arena):       786432
  Bytes in mapped regions (hblkhd):      21590016
  Total allocated space (uordblks):      684448
  Total free space (fordblks):           101984
  Topmost releasable block (keepcost):   99456
==============================================================
Packet I/O Totals:
   Received:           164
   Analyzed:            81 ( 49.390%)
    Dropped:             0 (  0.000%)
   Filtered:             0 (  0.000%)
Outstanding:            83 ( 50.610%)
   Injected:             0
==============================================================
Breakdown by protocol (includes rebuilt packets):
```

```
File  Edit  View  Terminal  Tabs  Help
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.037 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.060 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.037 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.039 ms

--- 127.0.0.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4095ms
rtt min/avg/max/mdev = 0.037/0.046/0.060/0.011 ms
labex:project/ $ curl http://127.0.0.1
curl: (7) Failed to connect to 127.0.0.1 port 80 after 0 ms: Con
labex:project/ $
labex:project/ $ ping -c 5 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.079 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.041 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.050 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.044 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.040 ms

--- 127.0.0.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4098ms
rtt min/avg/max/mdev = 0.040/0.050/0.079/0.014 ms
```

## Result

Thus, Snort IDS/IPS was successfully used to detect and prevent real-time network threats using the TryHackMe platform.
The experiment enhanced understanding of intrusion detection, prevention mechanisms, and real-time network security monitoring.